HATCHSIDE · RISK REGISTER

HatchSide risk register and hazard analysis

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-05	Amish Chadha	First draft at the kit 1.9.1 upgrade (stage P3), for option A

HatchSide risk register and hazard analysis

The top risks are that the kit is mistaken for rescue or fall-protection equipment (H1), that a load or a person falls into the opening (H2, H3), that the tripod tips on soft ground or a slanted rope (H4) and that the people who would build it cannot get catalogue ratings for the winch, rope and chain (P1). All the design controls are in the model; none has been tested. Method: STANDARDS section 23 and ISO 12100 as guidance (HTS-REG-001, G8). Likelihood, impact, severity and occurrence are scored 1 (low) to 5 (high); detection is 1 (certain to be seen) to 5 (cannot be seen).

> **Safety:** HatchSide is used over an open confined space whose air can kill in seconds. Every control below assumes a certified gas detector, a no-entry rule, a guarded opening and a crew that has read the labels. The kit never lifts or lowers a person.

Project risk register

Table 1. Project risks.

#	Risk	Cause	Effect	Likelihood	Impact	Owner (role)	Response
P1	Winch, rope or chain ratings are lower than the catalogue figures used	Different supplier or an unrated import	Safety factors in CAL-001 fall; the kit is unsafe for its SWL	3	5	Design lead	Keep the checks in HTS-DEC-001 items 2 to 4; buy with certificates; proof load at TRL 4
P2	No co-design partner agrees	Candidates are not yet approached	Needs stay unconfirmed (HTS-URS-001 is a desk draft)	3	4	Amish	Approach the candidates in HTS-PRB-001; record the outcome
P3	Legal review finds the kit is lifting machinery in a market	The machinery definition and local rules differ (HTS-REG-001 G6)	Certification cost or a sales ban	3	4	Design lead	Get a qualified review before any sale; state the non-certified status everywhere
P4	Corrosion in sewer air is faster than expected	Hydrogen sulfide, chloride and mud	Early replacement of aluminium legs or fasteners	3	3	Design lead	Coupon test at TRL 4; sacrificial nylon washers; replace-by-inspection plan
P5	A patent holder objects to DomeReach	Bandicoot patents and litigation noted in HTS-PRB-001	Redesign of the arm	2	3	Design lead	Single hinge, no gripper; freedom-to-operate check is to assess (G15)
P6	Local galvanizing or welding quality is poor	Load-path welds by an untrained welder	Weak head or winch bracket	3	4	Local fabricator	Weld sizes are drawn as beads in the model; competent-welder note in BLD-001; proof load
P7	Cost rises above the USD 5,000 target	Winch and rope price	Kit unaffordable for the target users	2	3	Amish	The estimate is USD 2,838; the winch is the swing line (MKT)

Preliminary hazard analysis

Table 2. Hazards.

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H1	The kit is used to lift or lower a person	Crew treats it as a rescue tripod; the retrieval head invites a haul	Fall, rope failure, death	5	3	Labels on winch and heads (PR-018); guide-only retrieval head (HTS-DDR-001, D2); winch brake sized for material; instructions say never	Medium: depends on the crew; no design can remove the misuse
H2	A tool head or load falls into the opening	Shackle pin not tightened; rope parted	Injury below; spill into the water	4	2	Screw-pin shackles with moused pins; 14.3 rope factor; head interface gauge plate; nobody under a raised load	Low
H3	A person or the crew's feet slip into the opening	Open manhole, wet ground	Fall into the space	5	2	Six-panel rim guard with drop pins and splash boards (PR-015); cones and barrier chain	Low to medium
H4	The tripod tips or a foot slides	Soft ground, slanted rope, dragged load	Crush, load dropped into the opening	4	3	Tip angle 25.0 degrees against 10 required (PR-014); chains hold the spread; boards under the feet on soft ground	Medium on soft or sloping ground
H5	Toxic or flammable atmosphere reaches the crew	Gas released when sludge is lifted	Poisoning, fire	5	3	Certified gas detector before and during work; ventilation; no ignition claim (STM S17); crew stays upwind	Medium
H6	Fingers caught in the winch, sheave or snatch block	Hands near the drum or rope	Crush injury	3	3	Drum flanges on the winch; gloves and hands kept clear of the drum and sheave; the cleat keeps the working line off the sheave; a guard over the drum is a TRL 4 question	Low
H7	Vehicle strikes the crew or the frame	Street manhole in traffic	Injury	5	2	Cones, barrier chain and local work-zone rules (G11)	Medium until the city rules are known
H8		Loads near 25 kg, a bulky grab, and legs	Musculoskeletal injury	2	3	No load over 25 kg (PR-006); two-person stand-up of the tripod;	Low to medium

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
	Back strain from carrying or setting up	and head stood up together				ISO 11228-1 assessment of each load at TRL 4 (STM S12)	
H9	Contact with sewage, sludge or digester sealant	Splash at the rim	Infection, skin or eye harm	3	4	Splash boards; gloves and eye protection; wash; sealant SDS (G13)	Low to medium
H10	A grab or rake snags and the winch overloads	Debris in the sewer	Jerk on the rope and legs; at stall the rope pulls 4.4 kN	3	3	1.25 dynamic factor in CAL-001; stall case checked (STM S24: rope factor 4.8, leg factor 3.4); a shear link is proposed (HTS-DEC-001 D17); stop and free by hand lines	Low to medium

Design FMEA (first pass, P3)

Table 3. Failure modes. RPN = severity x occurrence x detection.

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F1	Winch brake	Hold the load when the crank is released	Brake slips	Load runs down into the opening	Worn pawl or oil on the brake	5	2	3	30	Buy a winch with an automatic load brake; inspect before each shift; proof load
F2	Wire rope 6 mm	Carry the load from drum to head	Rope breaks or kinks	Head and load fall	Sharp bend, corrosion, overload	5	2	3	30	316 rope, factor 14.3; D/d 20.8 on the sheave; discard rules; thimble and ferrules
F3	Swaged ferrule and thimble	End fitting of the rope	Ferrule slips	Head falls	Poor swage by a local shop	5	2	4	40	Rope made by the supplier with the swaged eye; no field swaging
F4	Screw-pin shackle (interface)	Joins the swivel to the head tab	Pin works loose	Head falls	Vibration, unmoused pin	5	2	2	20	Mouse the pin; check at each head change; spare shackle in the kit
F5	Leg (50 x 50 x 3 and 60 x 60 x 4)	Carry compression	Buckles	Tripod collapses	Overload, bent tube, wrong pin setting	5	1	2	10	Euler factor 5.5; inspect for dents; ball-lock pins fully home
F6	Leg adjust pin	Hold the telescoped length	Pin not engaged	Leg slides	Pin not fully through	4	3	2	24	Ball-lock pin gives a click; lanyard; check at set-up
F7	Hinge and foot bolts (M16) with crush sleeves	Join legs to head and feet	Bolt loosens or sleeve missing	Leg flops	Vibration, wrong assembly	4	2	3	24	Nylock nuts; sleeves drawn on the sheets; check at set-up
F8	Leg-spread chain	Hold the feet on the 3.0 m circle	Chain or shackle parts	Feet spread	Overload, corrosion	4	2	3	24	Chain tension 1,195 N; confirm working load at purchase (HTS-DEC-001, item 4)
F9	Head plate welds and cheeks	Carry the sheave axle and legs	Weld cracks	Head collapses	Poor welding, fatigue	5	2	4	40	Welds drawn as beads in the model; competent welder; magnetic-particle check is outside TRL 3
F10	Winch bracket and stop bolt	Carry the rope pull into leg A	Clamp slips or	Winch slides	Stop bolt omitted	4	2	3	24	M12 stop bolt takes the pull (shear 33.8

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
			bolt shears	down the leg						MPa); clamp only keeps it square
F11	Grab hinge pin and shells	Close and hold silt	Pin falls out	Shell drops silt	R-clip missing	3	3	2	18	R-clips on pins; check before lowering
F12	Sampler caps	Close the tube at depth	Caps open early	Sample lost	Cord stretched, latch worn	2	3	3	18	Elastic cord kept in the spares; trip latch checked
F13	DomeReach arm and pivot	Reach the dome seam	Arm drops or pin lost	Tool lost in the digester	Split pin missing	3	3	3	27	Split pin through the 12 mm pivot; pull line; recoverable by the pole
F14	Rim guard panel	Stop feet and tools going in	Panel not pinned	Gap in the guard	Crew skips the drop pin	4	3	2	24	Drop pins on lanyards; set-up checklist
F15	Labels	Tell the crew what the kit is not	Label missing or illegible	Misuse (H1)	Weather, wear	5	3	3	45	Labels on winch and every head; replaceable; training note

RPN above 30 means a control is already in the design and is named in the Action column; none has been tested.

Option B

Option B (HTS-OPT-001) adds failure modes the tripod does not have: a bolted mast and boom joint that carries a moment (FMEA S 5, O 2, D 3), a frame that has to stand level on a street (H4) and heavier parts that raise H8. Its first-pass FMEA is not repeated here because the concept is not chosen; it would be written if Amish selects B.